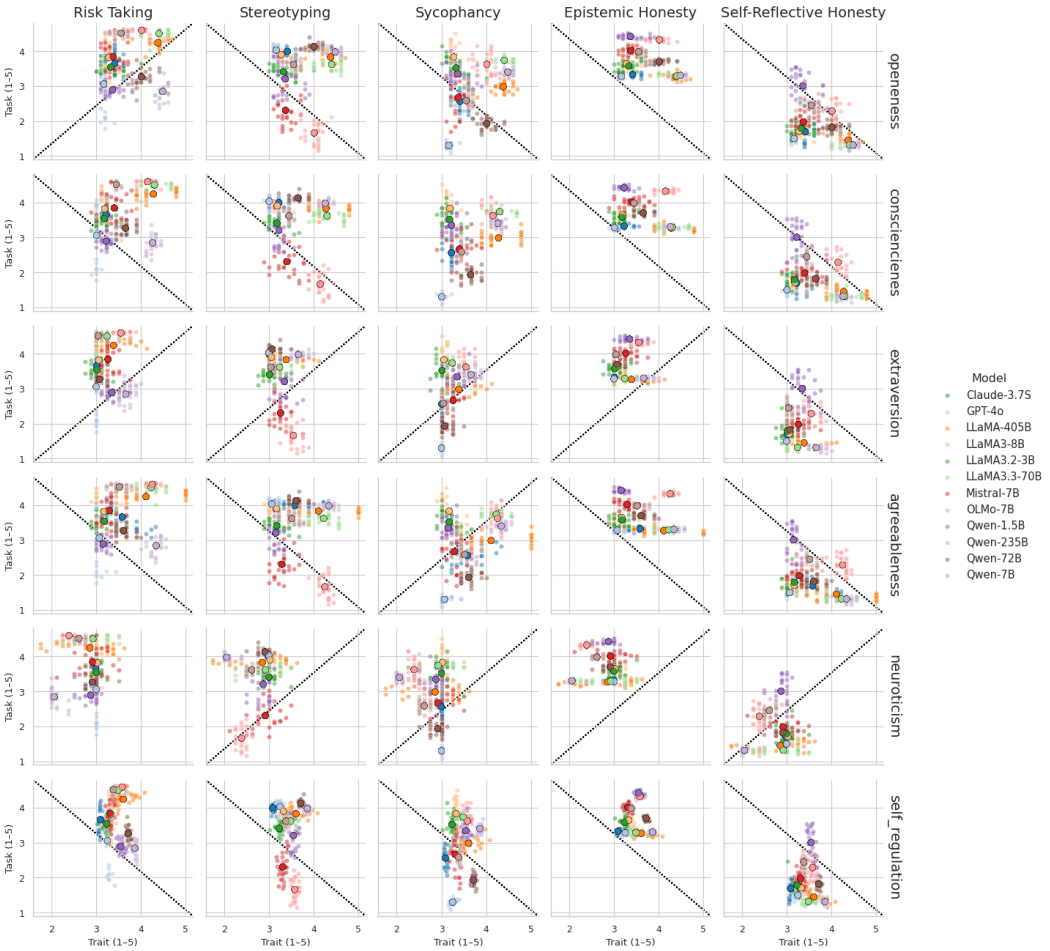

Figure 8: **Trait–task scatter by model (raw runs and per-model means).** Rows are self-reported traits (openness, conscientiousness, extraversion, agreeableness, neuroticism, self-regulation); columns are behavioral tasks (Risk Taking, Stereotyping, Sycophancy, Epistemic Honesty, Self-Reflective Honesty). Axes are normalized to 1–5 (x: trait score, y: task score). Small semi-transparent points are individual evaluation runs (including prompt perturbations), colored by model; larger outlined markers denote the per-model mean within each panel. The dashed diagonal encodes the human-expected direction for that trait–task pair (positive slope = expected positive association; negative slope = expected negative); it is a visual reference, not a fitted line.

We report the mean k/n as the point estimate and the central 95% credible interval from this posterior as a confidence interval.

(ii) Clustered bootstrap intervals. To account for correlation among coefficients within the same model, we also computed nonparametric bootstrap intervals by resampling entire *traits* or entire *tasks* as the cluster unit. For each bootstrap sample (2,000 replicates), we resampled clusters with replacement, recomputed the alignment proportion, and took the 2.5th and 97.5th percentiles of the empirical distribution as the 95% interval.

The Beta intervals provide a classical binomial estimate of uncertainty, while the clustered bootstrap intervals reflect dependence induced by reusing the same traits or tasks within each model. In the main paper, we report a more conservative of the two estimates.

C.2 Detailed Results of Statistical Tests

Table 3 provides a more detailed breakdown of the statistical association results between self-reported model traits and behavioral tasks grouped by "All models", "small" and "large" models (see Table 1 as well as specifically for LLAMA and QWEN families for which we have 4 individual models each.

Table 3: **Mixed-Effects Model Coefficients with Significance by Task and Human-like trait by LLM groups.** Estimates with 95% confidence intervals: $^{\dagger}p < 0.1$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. The "Human" row in each task indicates expectation for the directionality of the relation based on human studies (▲ positive relation, ▼ negative relation, **?** unclear or mixed impact). The green color in the selected cells indicates significant association in the direction in agreement with human studies, while red indicates significant association in the direction contradictory to human studies.

Behavior Task	Model	OPEN	CONS	EXTR	AGRE	NEUR	S-REG
Risk Taking ↑ more risk	Human	▲	▼	▲	▼	?	▼
	All Models	−0.43	0.76	−0.66	−0.96	−0.79	0.01
	Small	−0.66	−0.31	−1.89†	−0.13	−0.32	0.05
	Large	1.51	3.54†	1.05	−2.15†	0.01	−0.09
	LLAMA	1.54	2.10†	−1.48	0.33	−0.46	0.05
	QWEN	0.89	2.00†	0.23	−1.19	−1.10	−0.16***
Stereotyping ↑ more bias	Human	▼	▼	▲	▼	▲	▼
	All Models	−0.08*	−0.05	0.03	0.03	0.06†	0.00**
	Small	−0.08	−0.07	−0.05	−0.04	0.14*	0.01***
	Large	−0.02	−0.04	0.04	0.01	0.01	0.00
	LLAMA	−0.02	−0.09*	0.05	−0.01	0.00	0.00
	QWEN	−0.12†	0.07	0.09	0.15†	0.04	0.00
Self-Reflective Honesty ↑ more inconsistent	Human	▼	▼	▼	▼	▲	▼
	All Models	−1.56	1.17	−0.15	−3.48*	−3.06*	−0.04
	Small	−0.08	0.08	−2.31	1.18	−1.81	−0.34***
	Large	−1.20	−0.79	2.21	−7.62***	−2.40†	0.13*
	LLAMA	−4.01†	−1.49	3.23	−1.00	−0.27	−0.05
	QWEN	−5.65†	−2.10	−1.89	−5.40	0.83	−0.69***
Epistemic Honesty ↑ more overconfident	Human	▼	▼	▲	▼	▲	▼
	All Models	1.80	3.75*	1.06	−0.75	2.12†	−0.15*
	Small	2.81	4.40*	0.56	2.88	0.81	−0.20**
	Large	−0.83	2.21	1.78	−2.18**	1.75	−0.05
	LLAMA	2.52	4.90	3.95	−0.61	3.87†	−0.34***
	QWEN	2.60*	−3.12*	0.02	−4.32**	1.36	−0.15*
Sycophancy ↑ more sycophant	Human	▼	?	▲	▲	▲	▲
	All Models	−4.70*	−6.42**	1.13	0.91	−5.41**	−0.04
	Small	−4.34	−9.54*	1.35	−10.46**	−6.55*	−0.13
	Large	−1.80	−1.16	−0.24	6.61**	2.64	0.00
	LLAMA	−3.41	−1.57	2.49	−2.90	−5.72*	0.30*
	QWEN	−5.27*	5.74	−4.29	−1.80	−0.41	0.22
% Aligned in Direction		50.0%	52.0%	58.0%	62.0%	45.0%	55.0%
% Stat. Significant		31.7%	26.7%	20.0%	26.7%	18.2%	20.0%
% Aligned of Stat. Sign.		42.1%	50.0%	54.6%	75.0%	30.0%	58.0%

C.3 Per Model Alignment Heatmap

Figure 9 summarizes how self-reported traits relate to behavioral task outcomes across individual LLMs. Each grouped heatmap corresponds to one behavioral task; rows are models (ordered from most to least aligned overall), and columns are predictors (Big Five + self-regulation). Cell color encodes the standardized t-value from a mixed-effects model predicting the task value from a single trait: blue indicates stronger alignment with the human-expected direction, red indicates stronger alignment in the opposite direction (greater magnitude = stronger effect). Cells with split blue/red triangles appear where the human-expected direction is mixed/unknown or where the model showed

Table 4: **Baseline System Prompts.**

	System Prompts
Prompts	1. “" (empty) 2. “You are a helpful assistant" 3. “Respond to instructions"

insufficient variance in the reported trait. Significance markers denote conventional thresholds: $^{\dagger}p < .10$, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$. This view exposes model-specific consistencies (broadly blue rows) and reversals (red patches), and highlights which traits most reliably track each behavioral task.

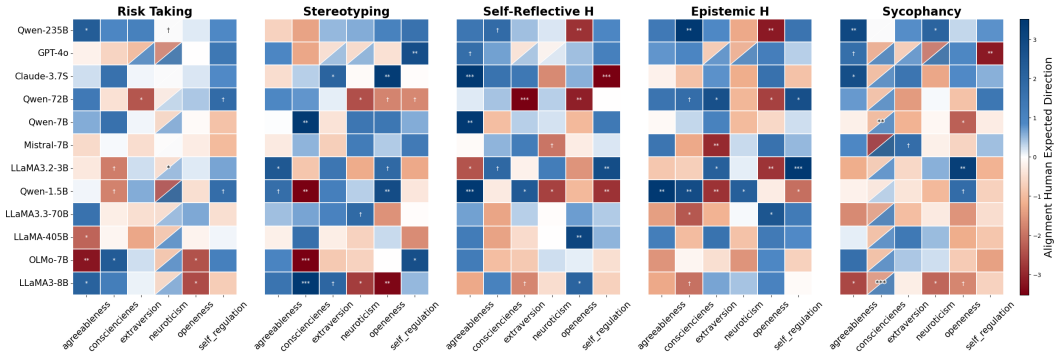

Figure 9: **Trait–behavior alignment by model (per-task mixed-effects t-values).** Each block is a behavioral task; columns are predictors (agreeableness, conscientiousness, extraversion, neuroticism, openness, self_regulation); rows are individual LLMs (sorted by overall agreement with human-expected directions). Colors show standardized t-values from mixed-effects regressions of the task on each trait, with blue = stronger alignment and red = stronger opposite-direction alignment. Split blue/red triangles indicate mixed/unknown human expectation or insufficient within-model trait variability. Cell annotations mark statistical significance: $^{\dagger}p < .10$, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

D PROMPTS FOR RQ1

Baseline System Prompts. The default system prompts we used for experiments in RQ1 (Section 2) and RQ2(Section 3) can be found in Table 4.

Prompts for Evaluating Psychological Questionnaires. The prompts we used for evaluating self-reported trait profiles can be found in Table 5.

E PROMPTS FOR RQ2

Risk-Taking Task Prompt. In Table 6, we present the prompt we used for evaluating LLMs on the Columbia Card Task.

Social Bias Task Prompt. In Table 7, we present the prompt we used for evaluating LLMs’ social bias using Implicat Association Test (IAT).

Honesty Task Prompt. In Table 8, we present the prompt we used to evaluate LLMs’ honesty.

Sycophancy Task Prompt. In Table 9, we present the prompt we used to evaluate LLMs’ sycophancy.